



DCDC Specifications

Models: TDC-JH

Products: 1.5KW OBC DC/DC converter

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1 Overview

1.1 Summary

The TDC-JH series 1.5KW vehicle DC converter is a product that provides vehicles with a 12V low-voltage DC power supply based on the DC national standard. The output end can be connected to the 12V reserve battery, and the DC/DC converter automatically charges the backup battery management. This product not only has the advantages of high efficiency, small volume, high stability, long life, etc., but also has the characteristics of high protection level, high reliability, and complete protection functions. It is an ideal power supply for electric vehicle charging. The product built -in thermal sensing device has an overheating protection function and can automatically recover. The full sealing process, the protection level of the IP67, can ensure that work in any complex environment will not cause failure.

1.2 Main features

1.2.1 Full -seal process, fan forced heat dissipation

1.2.2 The built -in temperature sensor can turn off the output under dangerous operating conditions (internal 110 ° C)

1.2.3 Protection level reaches IP67, which can work safely under short -term water soaking working conditions

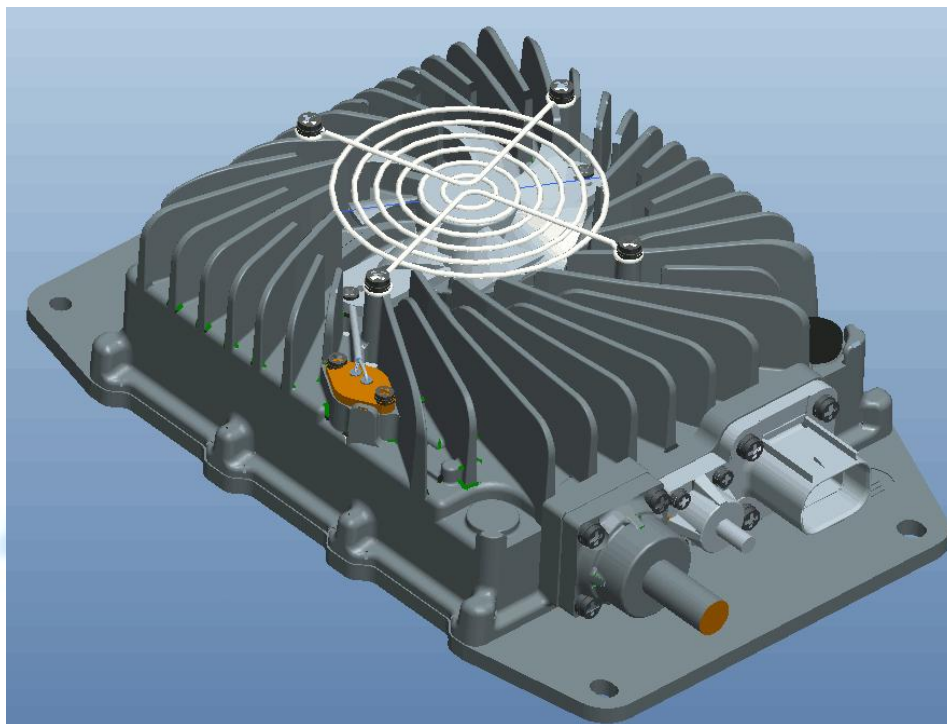
1.2.4 Compliance with CAN2.0 communication specifications, bus display work status, fault code, etc.

2.Size and Appearance

2.1 Size and Weight

Cooling Method	Length (mm)	Width (mm)	Height (mm)	Weight (KG)
Air cooling	256±1	183±1	79±2	3.5

2.2 Appearance



3. DC/DC converter technical indicator

3.1 Environmental requirements

▲ Working temperature

Area	Lowest temperature	Highest temperature
All over the world	-40℃	60℃

▲ Storage environment temperature

Area	Lowest temperature	Highest temperature
All over the world	-55℃	100℃

▲ Humidity: relative humidity 5%~ 95%, no condensation, no knot exposure

▲ Altitude: altitude ≤ 5000M

▲ Work noise: maximum noise at work ≤70dB, meet the GB-T24347-2009 standard

3.2 Regulations and reference standards for vehicle DC/DC converters

Items	Standard	Standard name	Notes
1	GB/T 24347-2009	Electric vehicle DC/DC converter	/
2	GB/T 18488.1-2015	Electric vehicle motor and its controller part 1: Technical conditions	/
3	GB/T 18384.2-2015	Electric vehicle safety requirements part 2: functional safety and fault protection	/
4	GB/T 18384.3-2015	Electric vehicle safety requirements Part 3: Personnel preventive electrical protection protection	/
5	GB/T 18387-2008	The limit and measurement method of the electromagnetic field launch intensity of the electromagnetic field of electric vehicles	/
6	GB/T 31498-2015	Safety requirements after the collision of electric vehicles	/
7	GB 9254-2008	The radio harassment limit and measurement method of information technology equipment	/
8	GB/T 18655-2010	Radio harassment of vehicle ships and internal combustion engines is used to protect the limit and measurement method for protecting the car receiver	/
9	GB 29743-2013	Motor vehicle engine coolant	/
10	GB 4208	GB 4208 shell protection level (IP code)	/
11	GB/T 28046-2	Environmental conditions and tests of road vehicle electrical and electronic equipment part 2: Electrical load	/
12	GB/T 28046-3	The environmental conditions and tests of road vehicle electrical and electronic equipment part 3: Mechanical load	/
13	GB/T 28046-4	The environmental conditions and tests of road vehicle electrical and electronic equipment part 4: Climate load	/
14	GB/T 2423.34-2012	Environmental Test Part 2: Test Method Test Z/AD: temperature/humidity combination cycle test	/
15	GB/T 2423.1-2008	Electrician Electronic Product Environment Test Part 1: Test Method B: Low temperature	/
16	GB/T 2423.2-2008	Electrician Electronic Product Environment Part 2: Test Method B: High temperature	/
17	GB/T 2423.3-2008	Electrician Electronic Product Environment Test Part 2: Test Method CAB: Constant Wet and Heating Test	/
18	GB/T 2423.17-2008	Electrician Electronic Product Environment Test Part 2: Test Method Test KA: Salt Mist	/

19	GB/T 30512-2014	Car disable material requirements	/
20	QC/T 413	Basic technical conditions for automobile electrical equipment	/

4.Car DC/DC converter security regulations

	Conditions	Standard
Ground resistance test	@25A/AC	$\leq 100\text{m}\Omega$
Input insulation test	@1000V/DC	$\geq 20\text{M}\Omega$
Enter resistance test	@2000V/DC 1min	Leakage current $\leq 10\text{ma}$

5.Electrical performance of the vehicle DC/DC converter

5.1 Input characteristics

Input	Rated voltage	DC72V	DC96V/ DC108V	DC144V	DC216V	DC320V
	Input voltage range	55~97V	73~137V	100~200V	170~300V	206~450V

5.2 Output characteristics

Output	Rated output voltage	14V				
	Output voltage range	9~15V				
	Rating output current	110A				
	Peak current	140A				
	Rated Power	1500W				
	Peak Power	1800W Last 6 minutes				
	Maximum efficiency	$\geq 90\%$	$\geq 92\%$	$\geq 92\%$	$\geq 93\%$	$\geq 94\%$
	Dynamic response time	$< 50\text{ms}$				
	Voltage adjustment	$\leq 1\%$				
	Load adjustment rate	$\leq 1\%$				
	Voltage control accuracy	$\leq 1\%$				
	Current control accuracy	$\leq 2\%$				

Quiescent Current	$\leq 1\text{mA} @ 14\text{V}$
Ripple voltage coefficient	$\leq 2\% @ \text{working status}$

5.3 Environmental test

Humidity test	Meet GB/T 24347-2009 6.1.2
Low temperature test	Meet GB/T 24347-2009 6.1.1.1.
High temperature test	Meet GB/T 24347-2009 6.1.1.2
Salt spray test	Meet GB/T 24347-2009 6.1.3
Electromagnetic	Meet GB/T 18487.3-2001 11.3.1.
Electromagnetic harassment	Meet GB/T 18487.3-2001 11.3.2
Protection level	IP67
Vibration	10 ~ 25Hz amplitude 1.2mm, 25-500Hz 30, 8 hours in each direction
M T B F	150000H

6.OBC DC/DC converter protection function

Protective function	Enter overvoltage protection					
	Enter under pressure protection					
	Output overvoltage protection	Over -voltage protection threshold of the output voltage is $16 \pm 0.5\text{V}$, and the voltage is $\leq 14 \pm 0.2\text{V}$ and returned to normal work				
	Output owed pressure protection	The output voltage overvoltage protection threshold is $7 \pm 1\text{V}$, and the voltage is $\geq 9 \pm 0.2\text{V}$.				
	Output overcurrent protection	The output current is greater than the peak current stop output				
	Premium protection	Internal temperature at 100°C began to reduce power, protect the shutdown at 110°C , and recover from self -recovery				
	Short circuit protection	Yes, self -recovery				

7. Interface

The interfaces of the vehicle DC/DC converter and the connection to the vehicle electrical and battery connection include low -voltage interfaces and high -voltage interfaces, low -voltage interfaces include CAN communication signal lines and DC/DC outputs, high -voltage interfaces are DC/DC input;

Selected by the manufacturer,

Note: In the high -voltage power distribution box, the high -voltage fuse of 1.5 times the maximum voltage/current inputs the DCDC input.

7.1 High -voltage input connection plug -in interface and definition

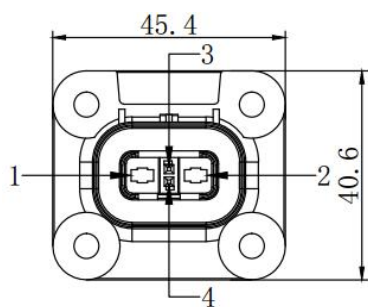


Figure 1 DC/DC input terminal

Product: 2103124-4		Client: 4-2103177-4	
Brand	Pin	Definition	Wire color and diameter
TE	1	Positive	red/2.5

	2	Negative	Black /2.5
	3	High -pressure interoperability signal1	Grey /0.5
	4	High -pressure interoperability signal 2	Brown /0.5

7.2 Signal connection interface and definition

7.2.1 Signal connection form 1:

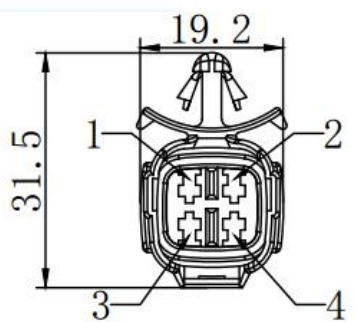
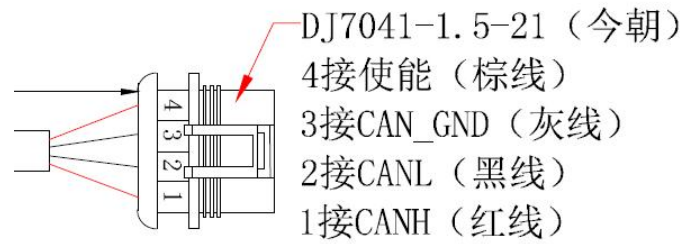


Figure 2 DC/DC signal side

Product : PP0427303		Clien: PP0144904	
Brand	Pin	Definition	Wire color and diameter
Tianhai	1	enable	blue/0.5
	2	Fault signal output	Yellow /0.5
	3	High -pressure interoperability signal 1	Grey /0.5
	4	High -pressure interoperability signal 2	Brown /0.5

7.2.2 Signal connection form 2:



Prdocut : DJ7041-1.5-21		Client : DJ7041-1.5-11	
Brand	Pin	Definition	Wire color and diameter
天海	1	CAN/H	Red /0.5
	2	CAN/L	Black /0.5
	3	CAN/GND	Grey /0.5
	4	enable	Brown /0.5

7.3 Low -voltage output interface and definition



Product : M6 nut		Client: M6 screw	
Brand	Pin	Definition	Wire color and diameter
TC	DC output	14V	Red/25

8. Security guide



Warning: Remind the user

- * It is strictly forbidden to disassemble and modify this device for repairs or debugging without authorization
- * Do not place the parts in the rainy position
- * Please confirm that the shell is intact before installation. If it is damaged, please change or contact after -sales
- * The plugs and sockets should be connected to fastener. If there is damage, loosen, please replace it immediately
- * It is strictly forbidden to carry out the connector in the working state of the product, otherwise it may cause personal damage
- * It is strictly forbidden to open the product shell in the process of product power, otherwise it may cause personal injury
- * It is strictly forbidden to touch the high -voltage band power parts of the naked hand. Please wear insulating gloves, insulating shoes, Insulation service, strictly forbidden power maintenance, testing
- * During the process of replacing the fuse and contactor, the barbaric operation is strictly prohibited, so as not to destroy the product's hidden safety hazards
- * AC power supply should choose the three -core cable with the ground wire and install the ground wire correctly

If there is an abnormal sound or odor during the working process, please unplug the power plug

When the battery is charged normally, please stay away from the fire source and flammable and explosive items

- * Do not charge the damaged or unsatisfactory battery

Note: Remind users that the following operations are important operations of this product

- * Do not block the product's air inlets and outlets to prevent overheating



* Please confirm that the output cable should not be too long, to avoid the effect of line voltage on charging

When mobile charging this product, please disconnect the power cord and charging plug

* The battery voltage must be consistent with the nominal voltage of the charger

* Avoid collision, pressure, not to pull, twist or shake charging cables

* The product should be placed in a safe, ventilated, dust -free, rainless environment

* It long -term do not use it for a long time, please pack and store

